Problem 7

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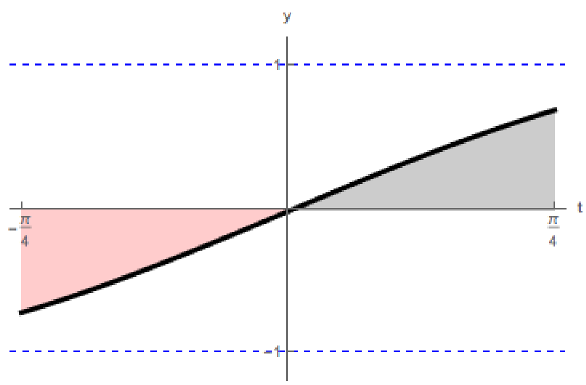
Problem 1 *Evaluate the following integrals using symmetry arguments.*

(a) $\displaystyle\int_{-\pi/4}^{\pi/4} \sin(t)\,dt$

Explanation. Since the function $f(t) = \sin(t)$ is odd, $\displaystyle\int_{-\pi/4}^{\pi/4} \sin(t)\,dt = 0$

We can illustrate our computation with the graph of the function f, where

$f(t) = \sin t$, for $-\dfrac{\pi}{4} \leq x \leq \dfrac{\pi}{4}$.



(b) $\displaystyle\int_{-2}^{2} (1 + x + 3x^7 - x^9)\,dx$

Explanation. In order to take advantage of symmetry, we will split the given polynomial into a sum of its even and odd parts. We will then integrate each part separately:

$$\int_{-2}^{2} (1 + x + 3x^7 - x^9)\,dx = \int_{-2}^{2} 1\,dx + \int_{-2}^{2} (x + 3x^7 - x^9)\,dx$$

We will first integrate an even function:
$\displaystyle\int_{-2}^{2} 1\,dx = 2\int_{0}^{2} 1\,dx = 2(2) = 4$

Then an odd function:
$\displaystyle\int_{-2}^{2} (x + 3x^7 - x^9)\, dx = 0$

Therefore: $\displaystyle\int_{-2}^{2} (1 + x + 3x^2 - x^9)\, dx = 4$

(c) $\displaystyle\int_{-\pi}^{\pi} x\cos(x)\, dx$

Explanation. $\displaystyle\int_{-\pi}^{\pi} x\cos(x)\, dx = 0$, since the product of an odd and even function is odd.

We can illustrate our computation with the graph of the function f, where

$f(x) = x\cos x$, for $-\pi \leq x \leq \pi$.